12 Feb 2026

Statement of Work (SOW)
MODIFY VALVE TEST STAND, X-31I

1. General

a. The contractor shall conduct modification of an existing valve test stand (hereinafter referred to as "the equipment") located at X-31I Inside Machine Shop, Bldg. A-48, Ship Repair Facility and Japan Regional Maintenance Center (SRF-JRMC), US Naval Base Yokosuka in accordance with this SOW.

b. The modification is required to improve usability of the equipment. The scope also includes preventive maintenance.

c. Unless otherwise specified herein, the contractor shall furnish all labor, materials, equipment, transportation, and supervision necessary to accomplish the work.

d. Utilities that are connected to the equipment
(1) Compressed air, 100 psig
(2) City water

e. The contractor shall have a local Japanese dealer network that provides maintenance services after modification.

2. Reference

The contractor shall refer to the following standard documents to accomplish the work to the extent specified herein.
a. 29 CFR 1910- Occupational Safety and Health Standards

3. Enclosure

(1) Concept Sketch of External Unit
(2) Installation Site
(3) Existing Air Piping

4. Existing equipment

Valve test stand
Mfr: Yamamoto Suiatu Kogyosho Co., Ltd.
Model: 1TA-16-5010
Serial Number: 1TA165010
Year of manufacture: 2016

5. Minimum Requirements

a. Unless otherwise specified, stated values represent the minimum.

b. Surfaces in contact with the test water shall be oil-free.

c. The modification shall be conducted in accordance with technical instruction from the manufacturer.

5-1. Modification of the existing equipment

a. Replace an existing air-driven water pump (Model: 5L-SS-30, Mfr: Hydraulics International) with a new one that meets the following specifications.

(1) Pressure ratio: 8±1

(2) Outlet pressure: 1,100 psig

(3) Volume displacement per cycle: 13 cm^3

(4) Proposal model: 3L-SS-8 (Mfr: Hydraulics International, Inc.) or equivalent

b. Replace an existing air regulator (Model: SMC, Model: AR40-N04H-Y-B) installed at the primary-side of the air-driven water pump with new one.

Range: 10 psig to 100 psig

c. Replace an existing relief valve (Mfr: Butech, Model: 5RVHS, Set pressure: 3,500 psig) with new one set to an appropriate pressure as recommended by the pump manufacturer.

d. Replacement of pressure gauge and piping modification

(1) Replace an existing pressure gauge that indicates hydro test pressure with new one that meets the following specifications.

(a) Range: 0 to 150 psig

(b) Indicator size: Φ 100 mm

(2) Modify piping so that the new pressure gauge indicates pressure of the driving air of the new air-driven water pump.

(3) Label the new pressure gauge as “Driving Air Pressure”.

e. Branch an existing city water inlet piping in the enclosure of the equipment. Add a new city water outlet port to supply city water to the external unit.

f. Branch an existing air inlet piping in the enclosure of the equipment. Add a new air outlet port to supply air to the external unit.

g. Branch an existing test media piping in the enclosure of the equipment. Add a new test media port to receive pressurized water from the external unit. Nominal pipe size shall be 1/8 inch. Allowable pressure shall be greater than 3,500 psig.

5-2. Fabrication of External Unit

a. The contractor shall design and fabricate an external unit in accordance with Enclosure (1).

b. The external unit shall be connected to the equipment as follows.

(1) City water and compressed air shall be supplied from the new ports of the equipment fabricated in accordance with Paragraph 5-1.e and 5-1.f.

(2) Water pressurized by the external unit shall be returned to the new test media port of the equipment fabricated in accordance with Paragraph 5-1.g.

(3) Piping shall be designed to match with system pressure and flow rate.

(4) Piping shall be firmly fixed to the floor.
(5) Valves shall be installed at both ends of each piping.

c. The following air-driven pump shall be installed.
(1) Pressure ratio: 25±1
(2) Maximum allowable pressure: 3,500 psig
(3) Volume displacement per cycle: 4.0 cm^3
(4) Proposal model: 3L-SS-25 (Mfr: Hydraulics International, Inc.) or equivalent
(5) The air-driven pump shall pressurize water supplied from the equipment.
(6) An air regulator shall be installed to adjust the outlet pressure within 500 to 3,500 psig.
(7) A speed controller shall be installed to adjust pumping speed.
(8) The following pressure gauge shall be installed between the air regulator and the pump.
(a) Range: 0 to 150 psig
(b) Indicator size: Φ 100 mm
(c) The gauge shall be labeled as “Drive Air Pressure”.

d. A lever-type hand pump shall be installed as follows.
(1) Outlet pressure: 1,000 psig
(2) The hand pump shall pressurize water supplied from the equipment.
(3) The installation height of the hand pump shall be determined so that a standing operator can operate it without difficulty.
(4) The hand pump shall be fixed firmly to a rigid steel frame.

e. A means shall be provided to switch the air-driven pump and the hand pump. Both pumps shall not be operated at the same time.

f. Check valves and relief valves shall be provided adequately to protect each component.

g. Depressurizing valves shall be provided for water and air systems. Water drain port shall be provided.

h. Components shall be fully enclosed.

i. Dimensions: 900 mm(W) x 700 mm(D) x 1,000 mm(H) or smaller

j. The external unit shall be able to be lifted by crane.

k. The external unit shall be able to be anchored on the floor.

l. Silencers shall be installed at all air exhaust ports.

5-3. Maintenance of air piping of the equipment

a. Reroute existing air piping so that it does not run over an existing H steel shown in Enclosure (3). The contractor may make a hole or slit in the H steel.

b. Replace the following parts shown in Enclosure (3) with new ones. Equivalent products are acceptable.

#	Nomenclature	Mfr	Model, description	Qty
1	Air cylinder	SMC	MBB100TN-295+295Z-XC10+Y-10M, Double-headed	2 ea
2	Speed controller	SMC	SMCAS4201F-N04-13SA	8 ea
3	Tube connector	Not specified	Φ1/2", elbow, swivel	8 ea
4	Plastic tube	Not specified	Φ1/2"	20 m

6. Work Procedure

a. The contractor shall conduct modification of the existing valve test stand in accordance with Paragraph 5-1.

b. Unless otherwise specified, the contractor shall deliver, unload, and position the external unit.

(1). When crane lifting operations are required, the contractor shall submit a lifting plan compliant with NAVFAC P-307 at least ten (10) working days prior to the lifting operation. The lifting plan shall include, but not limited to, the following.

(a) Weight, shape, and dimensions of the lifted material(s)

(b) Safe Working Load (SWL) of eye bolts, if any. For a four-point suspension, the SWL per point shall be at least 50% of the total weight.

(c) Sketch of lifting

(d) List of required lifting gears

(2) If the lifting plan submitted by the contractor is reviewed and approved by a weight handling POC, the Government can provide overhead crane and operator for unloading and positioning.

c. The contractor shall fix the external unit in accordance with Enclosure (2).

d. The contractor shall connect the external unit and the existing valve test stand.

e. The contractor shall conduct maintenance in accordance with Paragraph 5-3.

7. Operational test

After the completion of the modification, the contractor shall conduct operational test to check the equipment works as intended. The operational test shall include, but not limited to, the following.

a. Air pressurization up to supply air pressure.

b. Water pressurization up to 1,100 psig by the new air-driven water pump installed in the equipment

c. Water pressurization up to 2,400 psig by the new air-driven water pump installed in the external unit

d. Water pressurization up to 1,000 psig by the hand pump installed in the external unit

e. Leakage check

f. Table tilting

8. Document submission

The contractor shall submit the following documents. Unless otherwise specified, the document shall be written in Japanese or English.

a. Pre-work document

The contractor shall submit the following documents to the Technical Point of Contact (TPOC) before the start of work, no later than twenty five (25) working days after contract award.

(1) Work schedule
(2) Items and procedure of the modification
(3) Drawings of the external unit
(4) Parts list

b. Completion report

The contractor shall submit the following completion report in English and Japanese within ten (10) working days from completion of the repair service.

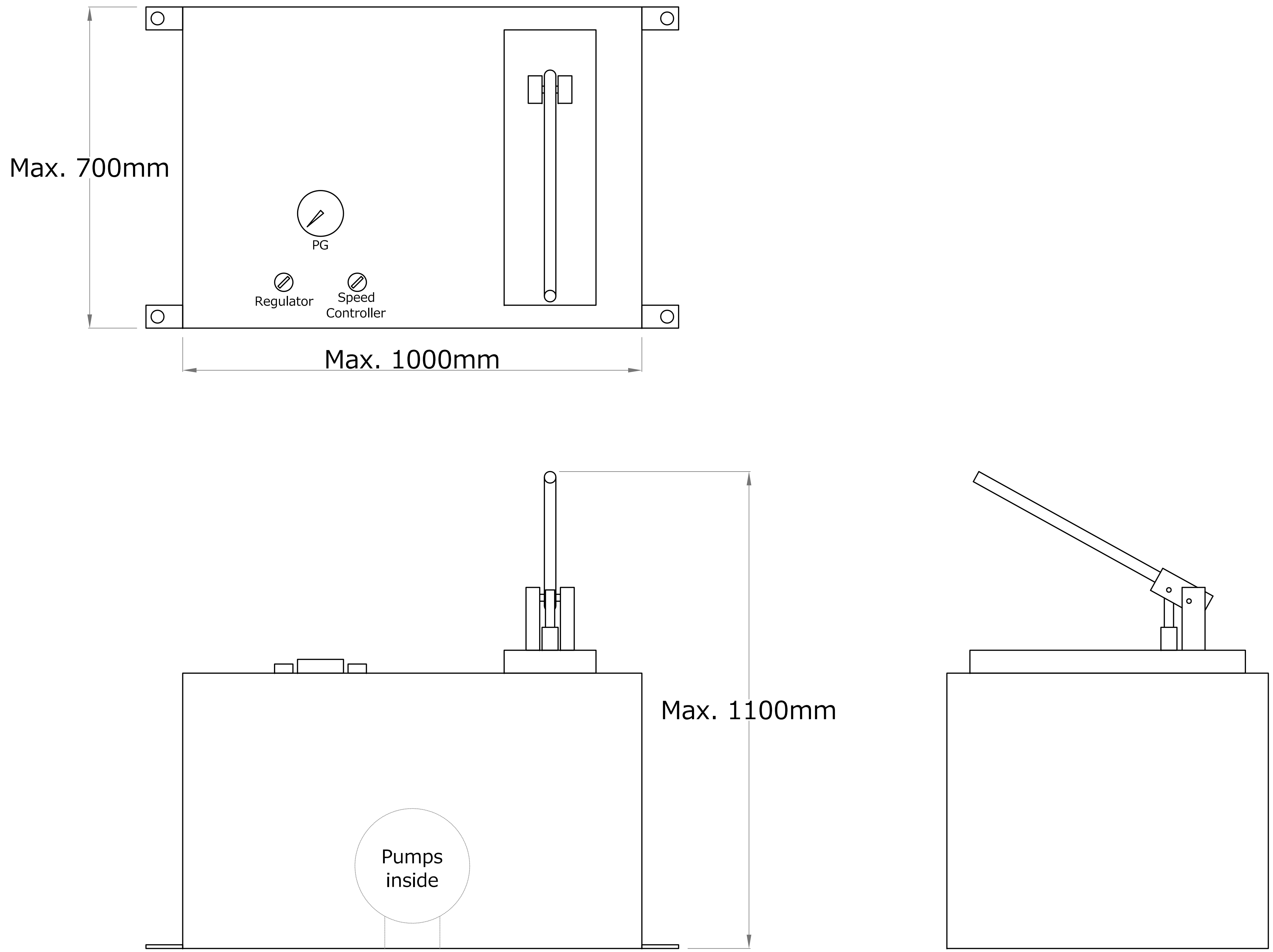
(1) Description of modification
(2) List of items provided
(3) Mechanical drawing and connection diagram
(4) Recommended date and items of the next maintenance
(5) SDS for all hazardous materials

c. If any unforeseen situation or damaged parts are found, the contractor shall issue condition report to the TPOC.

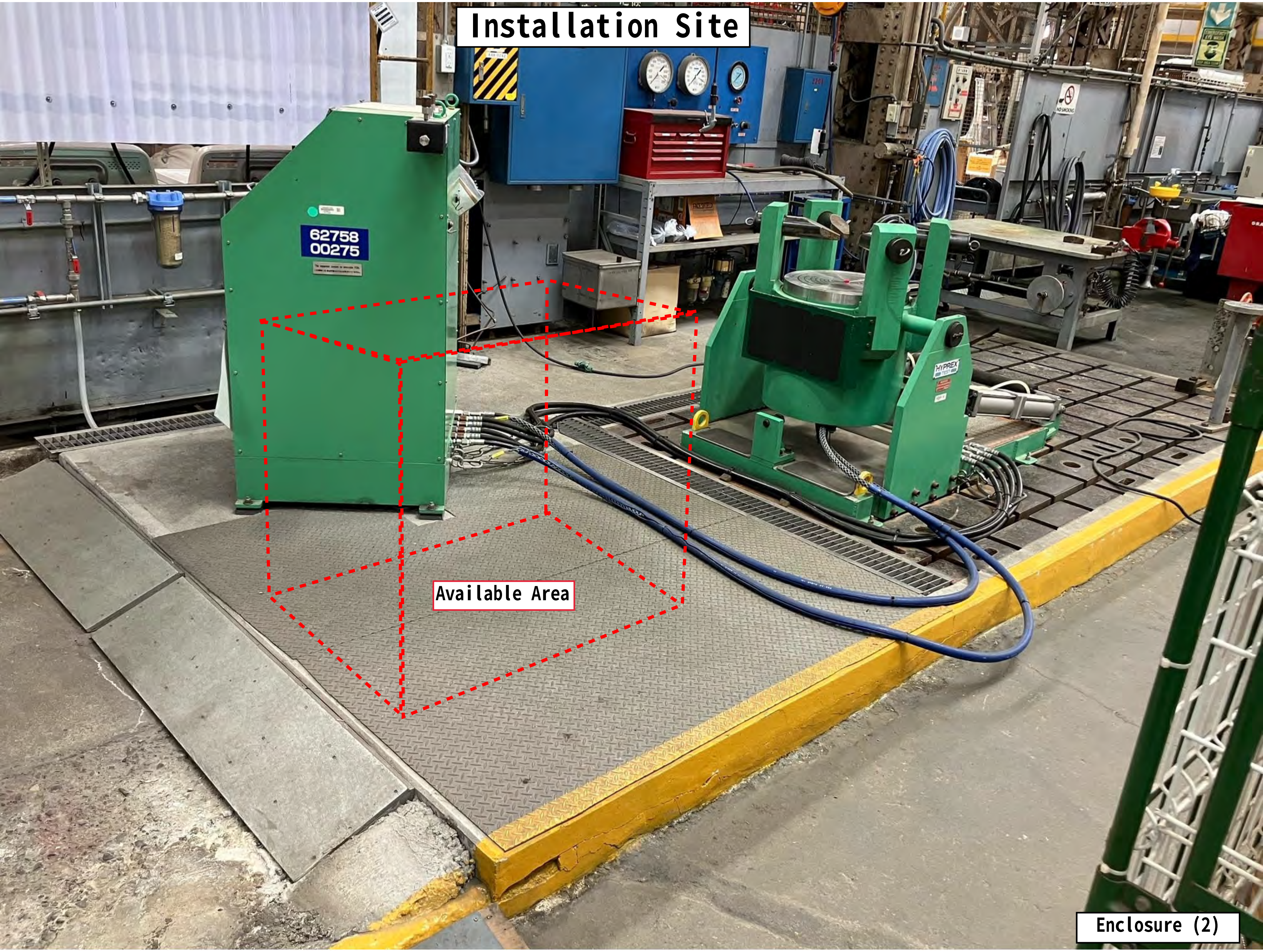
9. TPOC

KIMURA Kouroku, SRF-JRMC Yokosuka, C981
Phone: 046-816-4417
E-mail: kouroku.kimura.ln@us.navy.mil

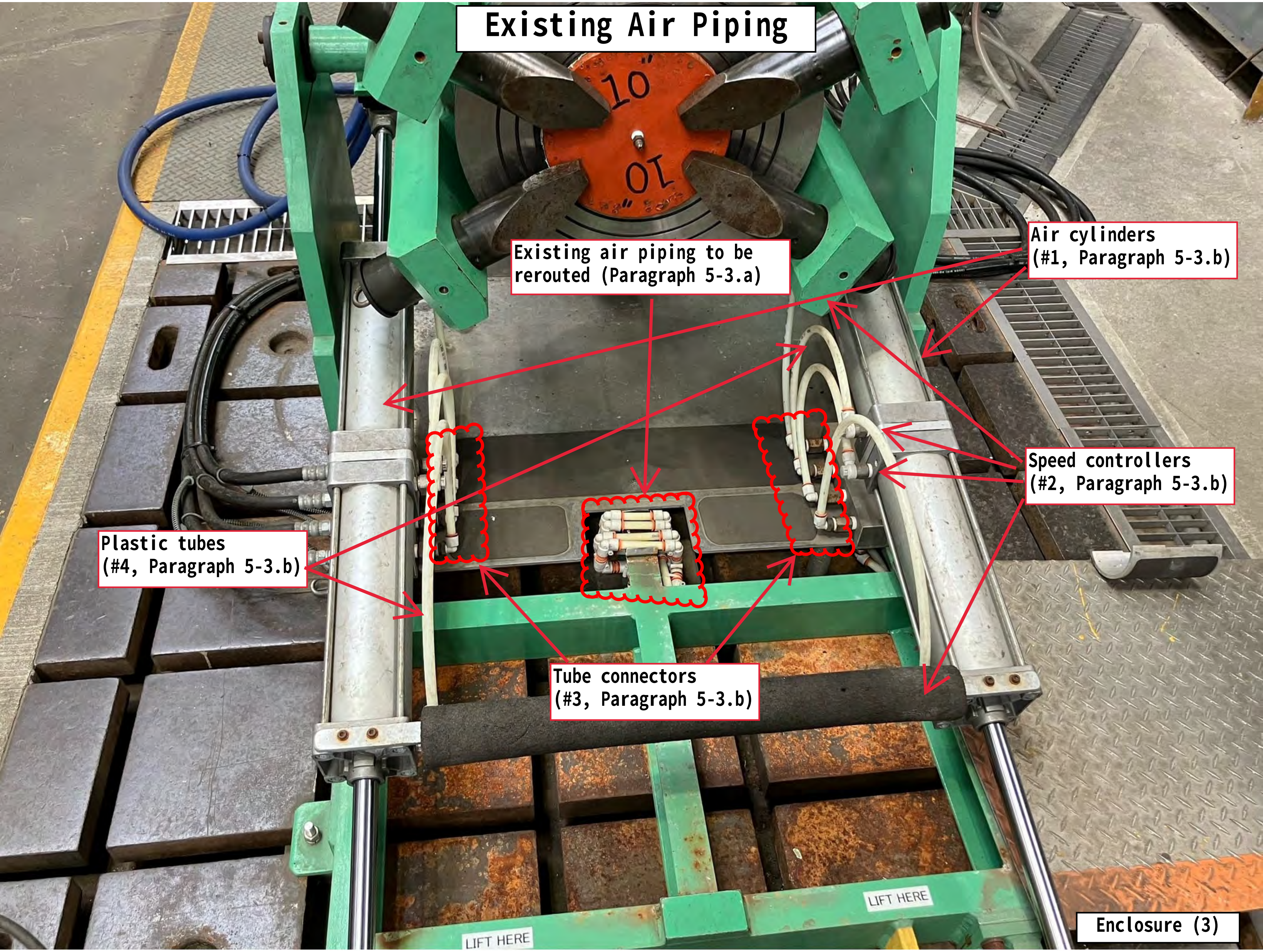
Concept Sketch of External Unit



Installation Site



Enclosure (2)


Existing Air Piping
Existing air piping to be rerouted (Paragraph 5-3.a)
Air cylinders (#1, Paragraph 5-3.b)
Speed controllers (#2, Paragraph 5-3.b)
Plastic tubes (#4, Paragraph 5-3.b)
Tube connectors (#3, Paragraph 5-3.b)
10"
10"
LIFT HERE
LIFT HERE
Enclosure (3)